



BHARAT ACADEMIX CODEQUEST · HACKATHON SUBMISSION

InternX AI

AI-Powered Virtual Internship Simulator

Technical Documentation · Hackathon Edition

Bridging the gap between academic learning and industry expectations through AI-generated internships, real project execution, and recruiter-ready credentials.

Team	Version	Date	Institution
CodeTitans	2.0 Hackathon Edition	June 2026	MSU

Prepared by CodeTitans

Executive Summary

InternX AI solves the experience paradox: students need jobs to get experience, but need experience to get jobs.

InternX AI is a production-grade, full-stack platform that simulates complete internship lifecycles — from AI-generated companies and managers to GitHub submissions, multi-dimensional AI evaluation, interview practice, and verifiable certificates — connecting students directly to recruiters and colleges.

4

User Portals

31+

MongoDB Models

19

API Route Groups

8

AI Engines

What We Built

- **Full-stack monorepo** — React 19 frontend + Express 5 backend + MongoDB Atlas
- **End-to-end student journey** — career selection → AI internship → tasks → GitHub → evaluation → certificate
- **Multi-tenant platform** — dedicated portals for students, colleges, recruiters, and admins
- **Groq AI integration** — 8 specialized AI engines with intelligent mock-data fallback
- **Production security** — JWT + RBAC + rate limiting + Helmet + Swagger docs

Team — CodeTitans

Dhruvil Dattani

Team Member · Maharaja Sayajirao University (MSU)

Dhruv Pathak

Team Member · Maharaja Sayajirao University (MSU)

Event: Bharat Academix CodeQuest | **Prepared by:** CodeTitans | **Version:** 2.0 Hackathon Edition

Table of Contents

— Hackathon Overview

★ Executive Summary

★ Why InternX Wins

1 Product Overview

2 Innovation — AI Engines

3 System Architecture

4 AI Ecosystem Architecture

5 Technology Stack

6 Complete InternX Workflow

6 Project Structure

7 Setup & Installation

— Frontend

8 Frontend Architecture & Flow

9 Landing Page Walkthrough

10 Routing & Route Guards

11 State Management

— Backend

12 Backend Architecture

13 API Endpoints Map

14 Database Models

— User Flows

15 Authentication Flow

16 Student Journey

17 College & Recruiter Portals

— Future

18 Future Roadmap

19 Build & Deployment

20 Key Files Reference

1. Product Overview

InternX AI is an AI-powered Virtual Internship Simulator that bridges the gap between academic learning and industry expectations.

Students gain real experience through AI-generated companies, projects, evaluations, certificates, and interview practice — before their first job.

Platform Actors

Role	Primary Features
Student	Career selection, AI internship, tasks, GitHub submission, AI evaluation, skill gap, interview simulator, certificates
College	Student management, placement tracking, analytics, certificate oversight, reports
Recruiter	Student discovery, shortlisting, hiring pipeline, contact requests, analytics
Admin	User approval, platform oversight, user inspection

2. Innovation — AI Engines

Five core AI innovations that differentiate InternX AI from every existing ed-tech or job platform.

AI Internship Generator

One-click generation of a complete virtual internship — AI company, manager persona, project brief, sprint tasks, and evaluation criteria — personalized to the student's chosen career path via Groq LLM.

Detail	Implementation
Backend	<code>internship.controller.js</code> → POST /api/internships/generate
Frontend	<code>InternshipGeneratorPage.jsx</code> , landing section <code>InternshipGenerator.jsx</code>
AI Model	Groq API / Llama 3.3 70B (with 7 career-path fallback mocks)
Output	Internship + Task documents in MongoDB

AI Manager & Mentor Agent

Simulated AI manager (Sarah Johnson archetype) assigns sprint tasks, provides contextual feedback on submissions, and mentors students through their internship journey — mimicking real workplace dynamics.

Detail	Implementation
Backend	<code>feedback.service.js</code> , <code>feedback.controller.js</code>
Frontend	<code>LivingAICompany.jsx</code> , <code>FeedbackCenterPage.jsx</code> , <code>AICompanyPage.jsx</code>
Features	Task assignment, progress tracking, personalized feedback messages

AI Evaluation Engine

Every GitHub submission is scored across five professional dimensions — Code Quality, Architecture, Performance, Security, and Documentation — producing objective, recruiter-

trusted assessment data.

Detail	Implementation
Backend	<code>evaluation.service.js</code> , <code>githubAnalyzer.service.js</code>
Frontend	<code>AIEvaluationPage.jsx</code> , <code>AIEvaluationEngine.jsx</code>
Models	<code>Evaluation.js</code> , <code>EvaluationReport.js</code> , <code>SkillGapReport.js</code>
Visualization	Recharts radar chart + ScoreRing components

AI Interview Simulator

Live AI-conducted mock interviews with dynamic question generation, real-time confidence/communication/technical scoring, and detailed post-interview reports.

Detail	Implementation
Backend	<code>interview.controller.js</code> → <code>/api/interview/*</code>
Frontend	<code>LiveInterviewPage.jsx</code> , <code>InterviewDashboardPage.jsx</code> , <code>InterviewSimulator.jsx</code>
Models	<code>Interview</code> , <code>InterviewQuestion</code> , <code>InterviewAnswer</code> , <code>InterviewReport</code>

Recruiter Intelligence

Recruiters discover students with verified skill scores, evaluation history, and certificates — shortlist candidates, manage hiring pipelines, and send contact requests through a dedicated portal.

Detail	Implementation
Backend	<code>modules/recruiter/</code> → <code>/api/recruiter/*</code>
Frontend	<code>StudentDiscoveryPage</code> , <code>ShortlistedCandidatesPage</code> , <code>HiringPipelinePage</code>
Models	<code>ShortlistedCandidate</code> , <code>HiringPipeline</code> , <code>ContactRequest</code> , <code>Offer</code>

3. System Architecture

InternX AI — AI-Powered Virtual Internship Simulator — Detailed System Architecture

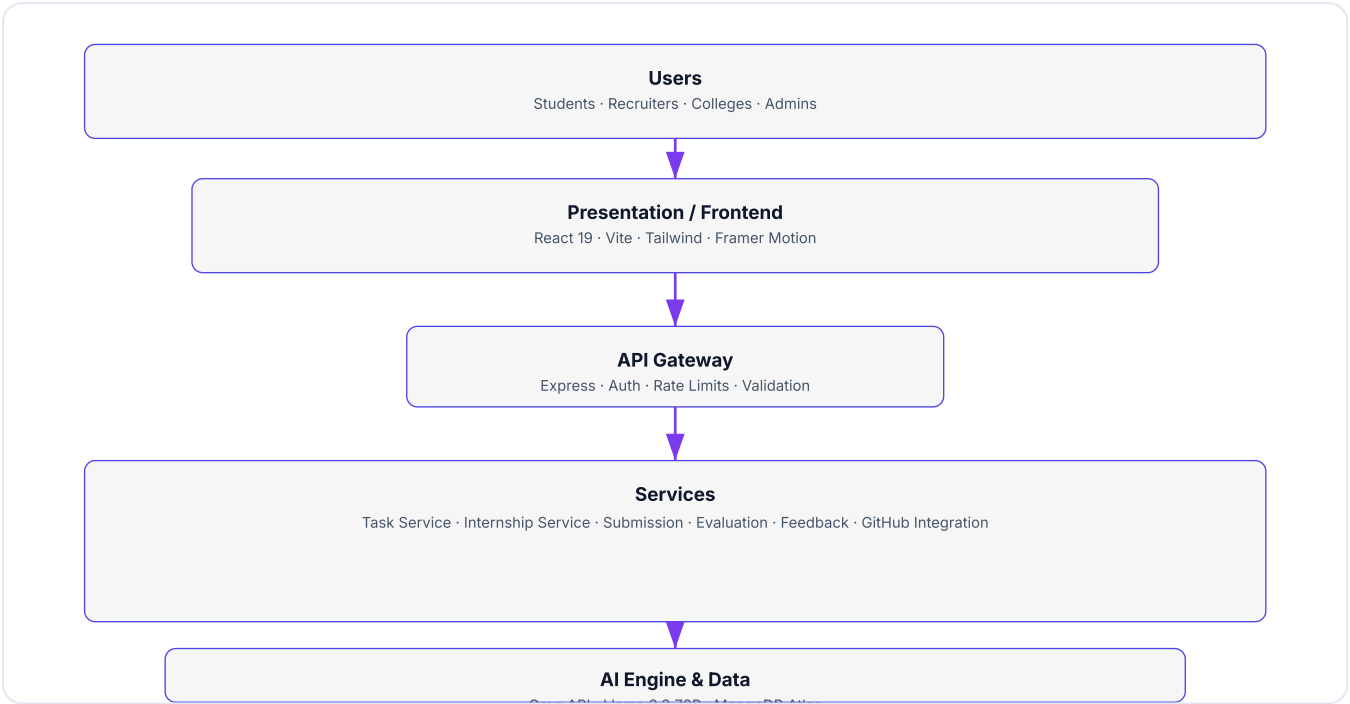


Figure 1 — InternX AI layered system architecture (Users → Presentation → API Gateway → Services → AI Engine & Data)

Architecture Layers (Implementation Mapping)

Users • Students — Learn · Build · Grow (career selection, internships, tasks, evaluation, certificates) • Recruiters — Discover · Evaluate · Hire (student discovery, shortlist, hiring pipeline) • Colleges — Monitor · Analyze · Guide (student management, placement, analytics, reports) • Administrators — Manage · Configure · Control (user approval, platform oversight)
Presentation Layer — React.js Frontend Application Path: frontend/src/ · React 19 + Vite 8 · Port 5173 • Tailwind CSS 4 — design tokens, glass UI, light/dark theme (index.css) • Redux Toolkit — github, college, recruiter, offers, notifications, placement slices • React Router 7 — HashRouter with lazy-loaded pages and role guards • Axios — JWT attach + auto-refresh on 401 (api/axios.js) • Framer Motion — page transitions, section animations, UI micro-interactions

- **Recharts** — radar charts, area/line charts on dashboards and landing sections
- **GSAP + ScrollTrigger** — horizontal product journey scroll, certificate reveal
- **Lenis** — smooth scroll ([hooks/useLenis.js](#))
- **React Context** — Auth, Theme, Navigation providers

API Gateway & Security Layer — Express.js API Gateway

Path: `backend/src/app.js` · Express 5 · Port 5000

- **JWT Authentication** — access token (15m) + HttpOnly refresh cookie (7d)
- **RBAC Authorization** — roles: student, recruiter, college_representative, admin
- **Rate Limiting** — 100 req/15min (10,000 in development) via express-rate-limit
- **Request Validation** — express-validator on auth and module routes
- **API Routing** — 17 REST route groups under `/api/`
- **CORS & Helmet** — CLIENT_URL whitelist, security headers, compression
- **Passport GitHub** — OAuth session for repository integration

Core Business Services

Service	Route / Module	Responsibility
Authentication Service	<code>/api/auth</code>	Register, login, OTP, refresh, profile, password reset
Career Path Service	<code>/api/careers</code> , <code>/api/career</code>	Career selection, career intelligence reports
Internship Service	<code>/api/internships</code>	AI internship generation, company & project setup
Virtual Company Service	Internship + <code>AICompanyPage</code>	AI manager, team, department simulation
Task Management Service	<code>/api/tasks</code>	Sprint tasks, kanban status, deadlines
Submission Service	<code>/api/submissions</code>	GitHub repo & file submissions
Evaluation Service	<code>/api/evaluation</code>	5-dimension AI code/project scoring
Certificate Service	<code>/api/college</code> + <code>/verify/:id</code>	Verified credentials, QR verification
Recruiter Service	<code>/api/recruiter</code>	Discovery, shortlist, pipeline, contact requests

Service	Route / Module	Responsibility
Analytics Service	/api/skills, college module	Skill gap, placement readiness, dashboards
Notification Service	/api/notifications	In-app and email notifications
Admin Service	/api/admin	User approval, inspection, platform control

AI & Intelligence Engine — Powered by Groq AI

Service: `backend/src/services/groq.service.js` · Models: llama-3.1-8b-instant, qwen/qwen3-32b

AI Engine	Backend File	Function
AI Internship Generator	internship.controller.js	Generates company, manager, project, tasks
AI Virtual Company Generator	internship.controller.js	Builds AI companies & team context
AI Project Manager Agent	task.controller.js, feedback.service.js	Assigns tasks & tracks progress
AI Mentor Agent	feedback.service.js, careerReport.service.js	Guides & mentors students
AI Code Reviewer	evaluation.service.js, githubAnalyzer.service.js	Reviews code & provides feedback
AI Feedback Engine	feedback.controller.js	Generates actionable manager feedback
Skill Gap Analyzer	skills.routes.js, SkillGapReport model	Identifies gaps & recommends skills
AI Interview Engine	interview.controller.js	Conducts mock interviews & scoring

Fallback: predefined mock data used when GROQ_API_KEY is not configured.

Data Layer — MongoDB Atlas (Cloud Database)

ORM: Mongoose 9 · Config: `backend/src/config/db.js`

Collection	Model File
Users	models/User.js
Students / Recruiters	models/Student.js, models/Recruiter.js
Career Paths	models/CareerPath.js, models/StudentCareer.js
Internships	models/Internship.js
Virtual Companies	Embedded in Internship model
Tasks	models/Task.js
Submissions	models/Submission.js, models/SubmissionRepository.js
Reviews / Evaluations	models/Evaluation.js, models/EvaluationReport.js
Feedback	models/Feedback.js
Skill Reports	models/SkillGapReport.js, models/SkillAnalysis.js
Interviews	models/Interview.js, InterviewQuestion, InterviewAnswer, InterviewReport
Certificates	modules/college/models/Certificate.js
Analytics & Logs	CollegeAnalytics, PlacementReport, Notification
GitHub Data	models/GithubProfile.js, models/GithubRepository.js
Hiring Pipeline	modules/recruiter/models/HiringPipeline.js, ShortlistedCandidate

External Services

Service	Integration	Usage
Groq AI	groq.service.js	LLM for internships, evaluation, interviews, career intel
GitHub API	github.service.js, passport-github2	OAuth, repo sync, code analysis
Cloudinary	config/cloudinary.js	Document & avatar uploads

Service	Integration	Usage
Email Service	utils/mailer.js (Nodemailer)	OTP, password reset, notifications
QR Code Generator	CertificateReveal.jsx (Lucide QrCode)	Certificate verification display
Payment Gateway	—	Planned (Future)

Data Flow Legend

- **Solid arrows** — Request / Response flow (Frontend ↔ API ↔ Services ↔ Database)
- **Dashed arrows** — AI / Data processing flow (Services ↔ Groq AI ↔ MongoDB)
- **Bidirectional arrows** — Two-way sync (GitHub API, notifications, real-time dashboard data)

Monorepo Layout

```
internX123/  
├── frontend/      # React SPA (port 5173)  
├── backend/       # Express API (port 5000)  
├── docs/          # Documentation & architecture assets  
└── README.md
```

4. AI Ecosystem Architecture

High-level AI ecosystem and business workflow — focus on modules, responsibilities, and how data flows between them.



Student Layer

Student actions and signals: registration, career selection, task completion, GitHub submissions, interview sessions.

Career Path Engine

Maps student profile to recommended career templates, seeds internship generation inputs.

AI Internship Generator

Generates virtual companies, manager personas, projects and sprint tasks tailored to chosen career path.

AI Manager Agent

Assigns tasks, provides contextual mentorship, conducts performance reviews and feedback conversations.

GitHub Integration Layer

Handles OAuth, repo syncing, commit capture, and provides repository artifacts to analyzers.

Submission Evaluation Engine

Code & project analysis across multiple dimensions producing structured evaluation reports for recruiters and students.

Feedback Intelligence Engine

Transforms evaluation output into manager-like feedback, coaching tips, and next-action items for students.

Skill Gap Analysis Engine

Maps observed skills to industry benchmarks, produces readiness scores and personalized learning recommendations.

AI Mock Interview Engine

Conducts technical, HR, and behavioral interviews; scores communication, confidence and technical competence.

Certification Engine

Generates verifiable certificates based on completion and thresholds from evaluation & interview reports.

Business Workflow Notes

- Primary flow is student → generate internship → complete tasks → submit to GitHub → evaluation → feedback → skill-gap & interview → certificate → recruiter discovery.
- AI modules communicate via structured JSON artifacts (internship spec, task payloads, evaluation reports, feedback transcripts) stored in the service layer and MongoDB.
- Focus is on human-observable business outcomes (readiness score, certificates, recruiter reports) rather than internal DB relationships.

5. Technology Stack

Frontend

React 19 Vite Tailwind CSS

Framer Motion GSAP React Router

Axios React Query / TanStack Query

Lucide Icons Recharts

Backend

Node.js Express.js JWT Authentication

bcryptjs cookie-parser Helmet

express-rate-limit Multer Cloudinary

GitHub API Integration REST APIs

Database

MongoDB Atlas Mongoose ODM

AI & Intelligence Layer

Groq API Llama 3.3 70B

AI Manager Agent AI Mentor Agent

AI Feedback Engine AI Task Generator

AI Mock Interview Agent

Skill Gap Analysis Engine

Career Recommendation Engine

GitHub Repository Analyzer

DevOps & Deployment

GitHub GitHub Actions Render

Vercel Cloudinary

Environment Variable Management

Analytics

Recharts Performance Analytics Engine

Placement Readiness Engine

6. Project Structure

Frontend (`frontend/src/`)

Directory	Purpose
pages/	Route-level page components (lazy-loaded)
pages/dashboards/	Role-specific dashboard shells
components/sections/	Landing page marketing sections
components/layout/	Navbar, Footer, NotificationsDrawer
components/ui/	Reusable UI (buttons, cards, rings, theme toggle)
context/	Auth, Theme, Navigation providers
store/	Redux slices (github, college, recruiter, etc.)
api/	Axios instance + API service modules
routes/	ProtectedRoute, PublicRoute, RoleRoute guards
hooks/	useAuth, useLenis, useMagnetic, etc.

Backend (`backend/src/`)

Directory	Purpose
controllers/	Request handlers (auth, internship, interview, etc.)
routes/	Express route definitions
models/	Mongoose schemas
services/	Business logic (Groq, evaluation, GitHub, auth)
middlewares/	Auth, role, error handling
modules/college/	College portal (controller, routes, models)
modules/recruiter/	Recruiter portal (discovery, pipeline)

Directory	Purpose
config/	DB, JWT, Cloudinary, Passport, Swagger

7. Setup & Installation

PREREQUISITES

- Node.js 18 or higher
- MongoDB (local or MongoDB Atlas)
- Optional: Groq API key, Cloudinary, GitHub OAuth, SMTP credentials

BACKEND SETUP

```
cd backend
npm install
# Create backend/.env (see Section 6)
npm run dev      # → http://localhost:5000
npm start        # Production mode
```

FRONTEND SETUP

```
cd frontend
npm install
# Create frontend/.env
npm run dev      # → http://localhost:5173
npm run build    # Production build → dist/
npm run preview  # Preview production build
```

VERIFY INSTALLATION

Check	URL
API Health	http://localhost:5000/health
API Docs (Swagger)	http://localhost:5000/api-docs
Landing Page	http://localhost:5173/#/
Groq Debug	http://localhost:5000/api/debug/groq

6. Environment Variables

Backend (`backend/.env`)

```
# Core
NODE_ENV=development
PORT=5000
MONGODB_URI=mongodb://localhost:27017/internx
CLIENT_URL=http://localhost:5173

# JWT Authentication
JWT_ACCESS_SECRET=your_access_secret
JWT_REFRESH_SECRET=your_refresh_secret
JWT_ACCESS_EXPIRE=15m
JWT_REFRESH_EXPIRE=7d
SESSION_SECRET=your_session_secret

# AI (optional – uses mock data if missing)
GROQ_API_KEY=your_groq_key

# Cloudinary (optional – file uploads)
CLOUDINARY_CLOUD_NAME=
CLOUDINARY_API_KEY=
CLOUDINARY_API_SECRET=

# Email (optional)
EMAIL_HOST=smtp.gmail.com
EMAIL_PORT=587
EMAIL_USER=
EMAIL_PASS=

# GitHub OAuth (optional)
GITHUB_CLIENT_ID=
GITHUB_CLIENT_SECRET=
GITHUB_CALLBACK_URL=http://localhost:5000/api/github/callback
```

Frontend (`frontend/.env`)

```
VITE_API_URL=http://localhost:5000
```

8. Frontend Architecture & Flow

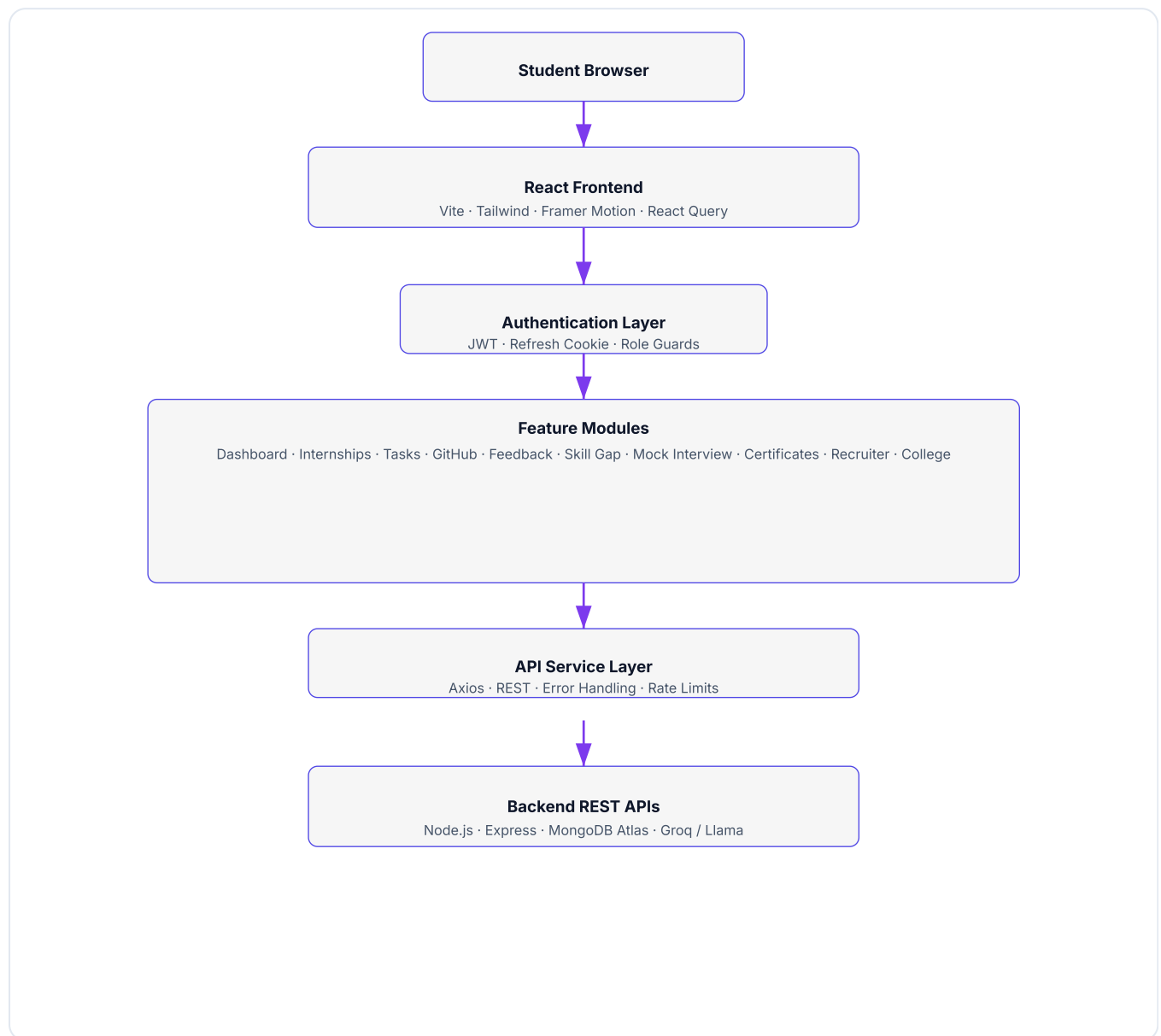


Figure 3 — Frontend application flow (student browser → frontend → auth → feature modules → API service → backend APIs)

Provider Stack (`main.jsx`)

```
Redux Provider
├── HashRouter
│   ├── AuthProvider
│   │   ├── ThemeProvider
│   │   │   ├── NavigationProvider
│   │   │   └── App
```

Global UI Layers (App.jsx)

- **Background canvas:** LiquidGlass, MatrixRain, SplashCursor, Spotlight
- **Layout:** Navbar, Footer (hidden on dashboard routes)
- **Overlays:** CommandPalette, NotificationsDrawer, ToastContainer
- **Transitions:** Framer Motion page fade/slide on route change
- **Loading:** PageSkeleton fallback for lazy-loaded routes

Code Splitting

All pages are lazy-loaded via `React.lazy()` + `Suspense` for optimal bundle size. Layout shells and route guards load eagerly.

8. Landing Page Walkthrough

Route: `/` → `LandingPage.jsx`

Scroll-driven marketing site with 12 animated sections:

#	Section	Component	Description
1	Hero	HeroExperience.jsx	Headline, CTAs, live dashboard mockup
2	Product Journey	ProductJourney.jsx	GSAP horizontal scroll — 7 steps
3	Internship Generator	InternshipGenerator.jsx	One-click AI company creation
4	Living AI Company	LivingAICompany.jsx	AI manager chat simulation
5	Project Execution	ProjectExecution.jsx	GitHub submission flow
6	AI Evaluation	AIEvaluationEngine.jsx	5-dimension radar scoring
7	Skill Evolution	SkillEvolution.jsx	Before/after skill morph on scroll
8	Interview Simulator	InterviewSimulator.jsx	AI interviewer + score rings
9	Certificate Reveal	CertificateReveal.jsx	GSAP certificate animation
10	Recruiter Experience	RecruiterExperience.jsx	Talent pool / shortlist UI
11	Bento Showcase	BentoShowcase.jsx	14-feature platform grid
12	Final CTA	FinalCTA.jsx	Conversion call-to-action

9. Routing & Route Guards

Guard	Behavior
PublicRoute	Login/register — redirects authenticated users away
ProtectedRoute	Requires authentication; shows PendingApprovalPage if unverified
RoleRoute	Restricts access by role: student, recruiter, college_representative, admin
RoleRedirect	Routes /dashboard to role-specific dashboard

Role → Dashboard Mapping

Role	Dashboard Route
student	/student/dashboard
college_representative	/college/dashboard
recruiter	/recruiter/dashboard
admin	/admin/dashboard

10. State Management

React Context

Context	Responsibility
AuthContext	Login, register, logout, getMe, JWT token management
ThemeContext	Light/dark theme toggle, persisted in localStorage
NavigationContext	Internship + tasks sync, toasts, student data fetching

Redux Slices

Slice	Domain
githubSlice	GitHub connection, repositories, profile
collegeSlice	College portal data
recruiterSlice	Discovery, shortlist, pipeline
offersSlice	Student job offers
notificationsSlice	In-app notifications
placementSlice	Placement tracking

11. Design System & Theming

Semantic CSS variables defined in `frontend/src/index.css` via Tailwind `@theme` :

Token	Usage
void, deep, surface	Background layers
text, muted, dim	Typography hierarchy
accent, violet, cyan	Brand colors & gradients
border, border-bright	Card & panel borders

Theme switching applies `.light` or `.dark` class on `<html>` . Custom utilities: `.glass` , `.glass-bright` , `.text-gradient` , `.grid-fine` , `.glow-accent` .

12. Backend Architecture

Middleware Pipeline

```
compression → helmet → cors → rateLimit → bodyParser
→ cookieParser → session → passport → routes → errorHandler
```

Security Features

- Helmet — HTTP security headers
- CORS — restricted to CLIENT_URL with credentials
- Rate limiting — 100 req/15min (10,000 in development)
- JWT access tokens (15min) + HttpOnly refresh cookies (7 days)
- bcrypt password hashing (salt rounds: 12)
- Role-based middleware on protected routes

13. API Endpoints Map

Prefix	Purpose
/api/auth	Register, login, logout, refresh, profile, password reset, OTP
/api/careers	Career path listing and selection
/api/career	Career intelligence reports (Groq-powered)
/api/internships	Generate, fetch, update AI internship
/api/tasks	Task CRUD, status updates, kanban
/api/submissions	Project code submissions
/api/evaluation	AI code/project evaluation (5 dimensions)
/api/feedback	AI manager feedback
/api/skills	Skill gap analysis and reports
/api/interview	Interview sessions, questions, live scoring, reports
/api/github	OAuth, repos, selected repository, code analysis
/api/student	Student-specific endpoints
/api/college	College portal — students, analytics, certificates
/api/colleges	College registry and lookup
/api/college-representative	College representative registration
/api/recruiter	Discovery, shortlist, pipeline, contact requests
/api/notifications	In-app and email notifications
/api/admin	User management, approvals, inspection
/health	Server health check
/api-docs	Swagger API documentation UI

14. Database Models

Model	Purpose
User	Core auth, role, profile, verification status
Student / Recruiter	Role-specific extended profiles
College	Institution registry
StudentCareer	Selected career path
Internship	AI-generated company, manager, project
Task	Sprint tasks with evaluation criteria
Submission	Code/project submissions
Evaluation / EvaluationReport	AI scores (5 dimensions)
Feedback	AI manager feedback messages
SkillGapReport	Skill analytics
CareerIntelligence	Career coach AI output
Interview / InterviewReport	Interview simulator data
GithubProfile / GithubRepository	GitHub integration
Notification / Offer / Placement	Comms & job tracking
Certificate	Verified credentials
ShortlistedCandidate / HiringPipeline	Recruiter module

15. AI Integration (Groq)

Service: `backend/src/services/groq.service.js`

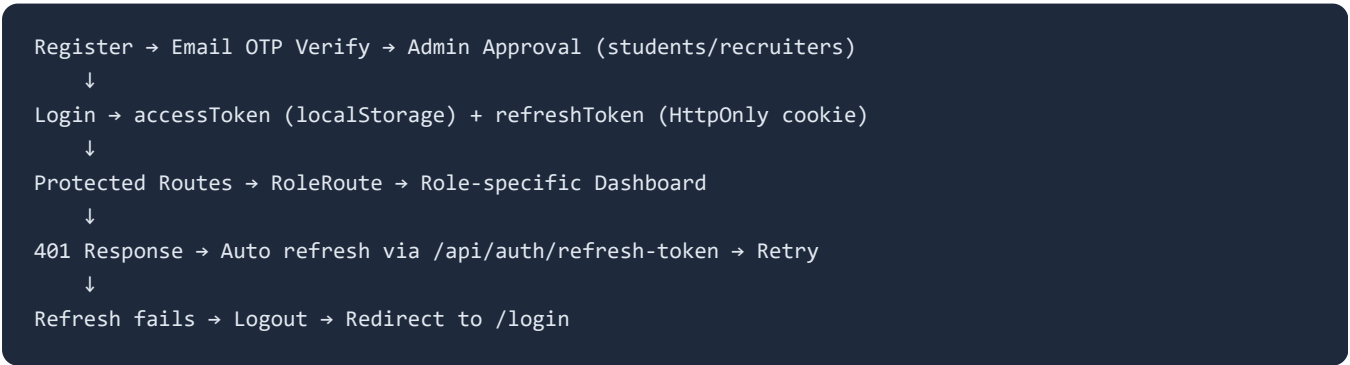
Models: llama-3.1-8b-instant, qwen/qwen3-32b

AI-Powered Features

Feature	What AI Generates
Internship Generator	Company, manager, project, tasks, culture
Evaluation Engine	Code quality, architecture, performance, security, docs scores
Interview Simulator	Dynamic questions, live scoring, reports
Career Coach	Personalized career intelligence reports
AI Manager	Feedback messages on task submissions

Fallback mode: If GROQ_API_KEY is not set, the platform uses predefined mock data — enabling full demo functionality without AI credentials.

16. Authentication Flow



Axios Token Refresh

The frontend `axios.js` interceptor queues failed requests during token refresh, preventing race conditions on concurrent 401 responses.

17. Student Journey (End-to-End)

Step	Route	API / Action
1. Onboarding	/onboarding	Profile setup
2. Career Selection	/career_selection	Select career path
3. Generate Internship	/generator	POST /api/internships/generate
4. AI Company Hub	/company	View manager, team, project
5. Task Kanban	/kanban	GET /api/tasks
6. Submit Work	/submit_task	GitHub / file submission
7. AI Evaluation	/evaluation	POST /api/evaluation/*
8. Feedback Center	/feedback_center	AI manager feedback
9. Skill Gap	/skill_gap	GET /api/skills/*
10. Analytics	/analytics	Performance dashboards
11. Career Coach	/career_coach	Groq career intel
12. Interview	/dashboard/interview	Live AI interview
13. Certificates	/certificates	Verified credentials
14. GitHub Module	/dashboard/github/*	OAuth + repo linking

17. College & Recruiter Portals

College Portal

Route	Feature
/college/dashboard	Overview & KPIs
/college/students	Student list & management
/college/students/:id	Individual student detail
/college/internships	Internship analytics
/college/skills	Skills analytics
/college/placement	Placement readiness
/college/certificates	Certificate management
/college/reports	Exportable reports

Recruiter Portal

Route	Feature
/recruiter/dashboard	Hiring overview
/recruiter/students	Student discovery & search
/recruiter/shortlisted	Shortlisted candidates
/recruiter/pipeline	Hiring pipeline stages
/recruiter/analytics	Hiring analytics
/candidate_profile	Shared candidate view

18. Future Roadmap

Post-hackathon vision for scaling InternX AI into a national internship infrastructure platform.

Phase	Timeline	Milestones
Phase 1 — Launch	Q3 2026	Payment gateway, mobile PWA, college bulk onboarding, 10 pilot universities
Phase 2 — Scale	Q4 2026	Multi-language support, voice-based interview simulator, LinkedIn integration, 100+ recruiter partners
Phase 3 — Intelligence	Q1 2027	Custom fine-tuned models, real-time collaboration rooms, AI pair-programming mentor, blockchain certificates
Phase 4 — Enterprise	Q2 2027	White-label for colleges, B2B recruiter SaaS, government skilling program integration, API marketplace
Phase 5 — Global	2027+	International career paths, cross-border recruiter network, industry-specific AI company templates

Vision: Make InternX AI the default pre-employment layer for every Indian graduate — turning the experience paradox into an experience advantage.

19. Build & Deployment

Frontend Production

```
cd frontend && npm run build    # → frontend/dist/  
# Deploy dist/ to Vercel, Netlify, or any static host
```

Backend Production

```
cd backend && npm start          # node server.js  
# Deploy to Railway, Render, VPS, etc.
```

Production Checklist

- Set NODE_ENV=production for secure cookies
- Set CLIENT_URL to production frontend URL
- Use MongoDB Atlas for production database
- Configure GROQ_API_KEY, Cloudinary, SMTP for full features
- HashRouter requires no server-side routing config
- Certificate links: /verify/:id redirects to hash route automatically

20. Key Files Reference

Concern	File Path
App routing	frontend/src/App.jsx
Auth context	frontend/src/context/AuthContext.jsx
Theme system	frontend/src/context/ThemeContext.jsx
Design tokens	frontend/src/index.css
Student state	frontend/src/context/NavigationContext.jsx
API client	frontend/src/api/axios.js
Internship AI	backend/src/controllers/internship.controller.js
Evaluation AI	backend/src/services/evaluation.service.js
Interview AI	backend/src/controllers/interview.controller.js
Groq service	backend/src/services/groq.service.js
Landing sections	frontend/src/components/sections/*
Route guards	frontend/src/routes/*
Express app	backend/src/app.js
Server entry	backend/server.js



InternX AI — AI-Powered Virtual Internship Simulator

Team CodeTitans · Bharat Academix CodeQuest · MSU

Technical Documentation v2.0 Hackathon Edition · June 2026

Prepared by CodeTitans · Dhruvil Dattani & Dhruv Pathak

React 19 + Express 5 + MongoDB Atlas + Groq AI

© InternX AI Platform · CodeTitans · Hackathon Submission